

3.8 Evaluation Design

Empirical evaluation assesses whether the three interaction modes—Ask, Code, and Agentic—behave according to their declared contracts when grounded in frozen Kaggle notebook snapshots. Experiments were executed with GLM 4.7 (`zai-glm-4.7`) via the Cerebras [API](#), using harness scripts under `testing/host/scripts/`. Each suite comprises four scenario tests aligned with mode-specific objectives identified in the literature on notebook assistants (McNutt & DeLine, [2023](#)) and tool-augmented language models (Yao et al., [2023](#); Qin et al., [2024](#)).

3.8.1 Benchmark suites

Ask mode measures explanation quality only: no code generation, no tool execution, and no notebook modification. Tests cover notebook workflow explanation, single-cell explanation, error diagnosis, and empty-cell handling. Metrics include topic coverage, evidence alignment, and a heuristic hallucination rate derived from snapshot ground truth.

Code mode measures code-generation quality in chat: runnable python blocks, placement guidance, and run-order instructions, without browser writes. Metrics include code-correctness (keyword alignment with notebook variables) and placement accuracy (cited cell indices relative to notebook flow).

Agentic mode measures *LLM tool dispatch* under the mandatory two-call contract. Each user task consumes exactly two [API](#) calls. In *round 0*, the model may issue read-only notebook queries such as `notebook_list_cells` and `notebook_get_cell`. In *round 1*, it emits the `implement batch`—including `insert_cell`, `edit_cell_by_index`, and `run_cell`—which the host dispatches in fire-and-forget mode without a closed-loop ReAct replanning cycle (Shinn et al., [2023](#); Yao et al., [2023](#)). Success is judged by dispatch fidelity of tool types, not by post-hoc execution verification (which the current artefact does not implement).

3.8.2 Notebooks and harness

Benchmarks used persistent JSON snapshots from public Kaggle kernels (Quaranta et al., [2021](#)). Primary fixtures were Pakistan housing (`kaggle_kernel_113620421.json`) and testing-ol (`kaggle_kernel_112732919.json`). The harness replays chat requests through `streaming.py` with browser tools mocked at the extension boundary. Trace logs are written to `agentic_tool_trace.jsonl`.